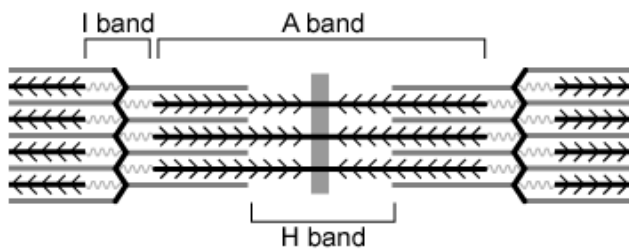


The diagram shows the positions of three different sections of a relaxed sarcomere.



During sarcomere contraction, which of the following would shorten in length?

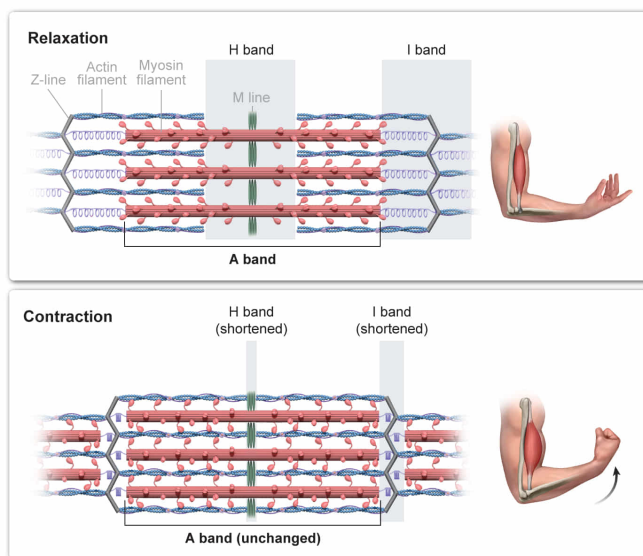
- ☐ A. A band only [10%]
- ☐ B. H band only [22%]
- ✓ ☒ C. I band and H band only [55%]
- ☐ D. A band, I band, and H band [11%]

Correct

55% answered correctly

Explanation:

The A band does not change in length during muscle contraction



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Skeletal muscle is muscle that is attached to bone and functions in voluntary movement. In this type of muscle, each **muscle fiber** (ie, muscle cell) contains thousands of **sarcomeres**, or individual functional units. Sarcomeres are crucial to the process of **muscle contraction**.

Sarcomeres consist of several sections:

- The **I band** consists of only actin (thin) filaments.
- The **H band** consists of only myosin (thick) filaments.
- The **A band**, which consists of both actin and myosin filaments, marks the length of the myosin filaments.
- The M line marks the center of each sarcomere.
- The Z line delineates the boundary between each sarcomere.

When muscle contraction occurs, the myosin and actin filaments interact. Specifically, myosin heads **bind** to the actin filaments and slide them toward the central point of the sarcomere. This interaction shortens the sarcomere's total length; however, the length of each actin and myosin filament remains *unchanged*. Instead, it is the sliding of the two filament types along one another that causes the sarcomere to shorten.

The question diagram shows the position of each of the three different sections in a relaxed sarcomere. During sarcomere contraction, the **lengths** of both the **I band** and **H band shorten** as the actin and myosin filaments slide along one another (**Choice B**). However, the A band, which corresponds to the length of the myosin filaments in one sarcomere, does *not* change in length (**Choices A and D**).

Educational objective:

In the sliding filament model, actin and myosin filaments slide across one another to shorten sarcomeres, causing contraction of skeletal muscle.

Time spent:0 sec

Subject:
Biology

QID:403204

System:
Musculoskeletal System